EEE407 Tutorial 2a

1. Define energy audit (Guidelines for conducting Energy Audit Commercial Buildings, Malaysia)

2. Draw the flow chart of a detailed energy audit process.

3. List five requirements of data collection.

4. Building A has a 6-level parking lot, T8 fluorescent lamp is used in every floor (see specification sheet:

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After performing field inspection, you have found out that there are total of 500 lamps being installed in each floor. An automated control system is used to turn on/off the lights at 3rd to 6th floor starting from 7pm to 7am the following day, whereas at 1st and 2nd floor, the lights are lit 24 hours due to no natural daylighting. Fill up the lighting data form and suggest zero cost ECMs to decrease the energy consumption. Based on your assumption, calculate the energy consumption after implementation of ECM.

Lighting Data Form

Desktop Data Collection

	Type of light:_____						
Level	Operation hours (hr/day)	Rated power kW (lamp + ballast)	Total no installed (nos)	Place of use	Control system (manual/auto)	Remark	Please expand the table for other type of lighting(s)

5. From question 4, the owner of the building would like to go for a medium cost ECM, which is to retrofit the existing lighting system with LED lamps. Find out the price of a typical LED variant of the same tube lamp, calculate the initial cost to purchase the LED lamps and a labour cost of RM2000. With tariff C1, how many months are needed to recoup back the initial cost.

3. Tariff C1 - Medium Voltage General Commercial Tariff		
For each kilowatt of maximum demand per month	**RM/kW**	**30.30**
For all kWh	**sen/kWh**	**36.50**
The Minimum Monthly Charge is	***RM***	***600.00***